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REFLECTION 1

Part 1

In this Data Mining course, I have gained a comprehensive understanding of foundational concepts essential for analyzing and extracting insights from complex datasets. The journey began with data types and attributes, where I learned to distinguish nominal (categorical), ordinal (ranked), interval (temperature), and ratio (height, weight) data, each with unique mathematical properties influencing preprocessing and analysis. Recognizing these distinctions ensures appropriate handling, such as avoiding invalid operations like multiplication on interval data.

A significant focus was on data quality challenges—noise, outliers, and missing values. Techniques like KNN imputation, statistical tests, and isolation forests address these issues, ensuring robust datasets. Preprocessing steps, including normalization, standardization, and feature engineering, emerged as critical for enhancing model performance. Feature selection methods like mRmR and Laplacian scores help reduce dimensionality while preserving relevance.

Distance metrics, such as Euclidean and Mahalanobis (accounting for feature correlations), and similarity measures (Jaccard, SMC) were highlighted as pivotal for algorithms like KNN and clustering. Understanding their assumptions—like feature independence in Euclidean distance—guides metric selection based on data characteristics.

Supervised learning methods, including decision trees (using entropy/Gini index for splits) and ensemble techniques (Random Forest, AdaBoost), illustrated the importance of combining models to reduce variance or bias. Unsupervised learning introduced clustering fundamentals. KNN's trade-offs between small/large *k* and distance metric choices underscored the balance between noise sensitivity and bias.

Overall, the course emphasizes a holistic workflow: ensuring data quality, preprocessing to highlight patterns, selecting appropriate algorithms, and validating results. These concepts collectively empower data-driven decision-making, blending theoretical principles with practical strategies for real-world applications.

Part 2

A. I attended almost every lecture in person as I find face-to-face interaction enhances my focus and facilitates immediate clarification of doubts. The structured environment minimizes distractions compared to remote learning. During COVID, while Zoom offered flexibility, I found face-to-face learning more effective due to direct engagement and real-time collaboration, which deepened my understanding of complex topics like neural networks and gradient boosting.

B. I reviewed slides and core reading materials before most classes, particularly for unfamiliar topics. However, I occasionally skimmed papers or book chapters if time permitted, prioritizing key concepts to actively participate during lectures.

C. Yes. After each week, I revisited algorithms by summarizing them in my own words and tested my grasp through coding exercises (e.g., implementing decision trees) or discussing nuances with peers. This practice helped solidify abstract concepts.

D. Absolutely. For unclear models, I supplemented lectures with online courses, research blogs, and GitHub repositories to see practical implementations and alternative explanations.

E. I am a Master's student in Computer Science. The algorithms are easy to moderate for me.

F. I had no difficulty understanding your speaking. Your clarity and pacing were effective.